

RM Search Plot Examples

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Setup

The following table lists the parameter values used to generate the plots shown in this document. The RM search was run as an array job, with the starting file index varied between runs to cover the full night of observations. The values used were [160,460,760,...,8860].

After the RM search, the pulse detection script (02_pulse_detection.py) was run as an array job with different FDF downsampling factors. This controls how much the FDFs are averaged before stepping through each time sample and detecting peaks along the phi axis. The downsampling factors tested were [2,4,6,8,10,12,14,16].

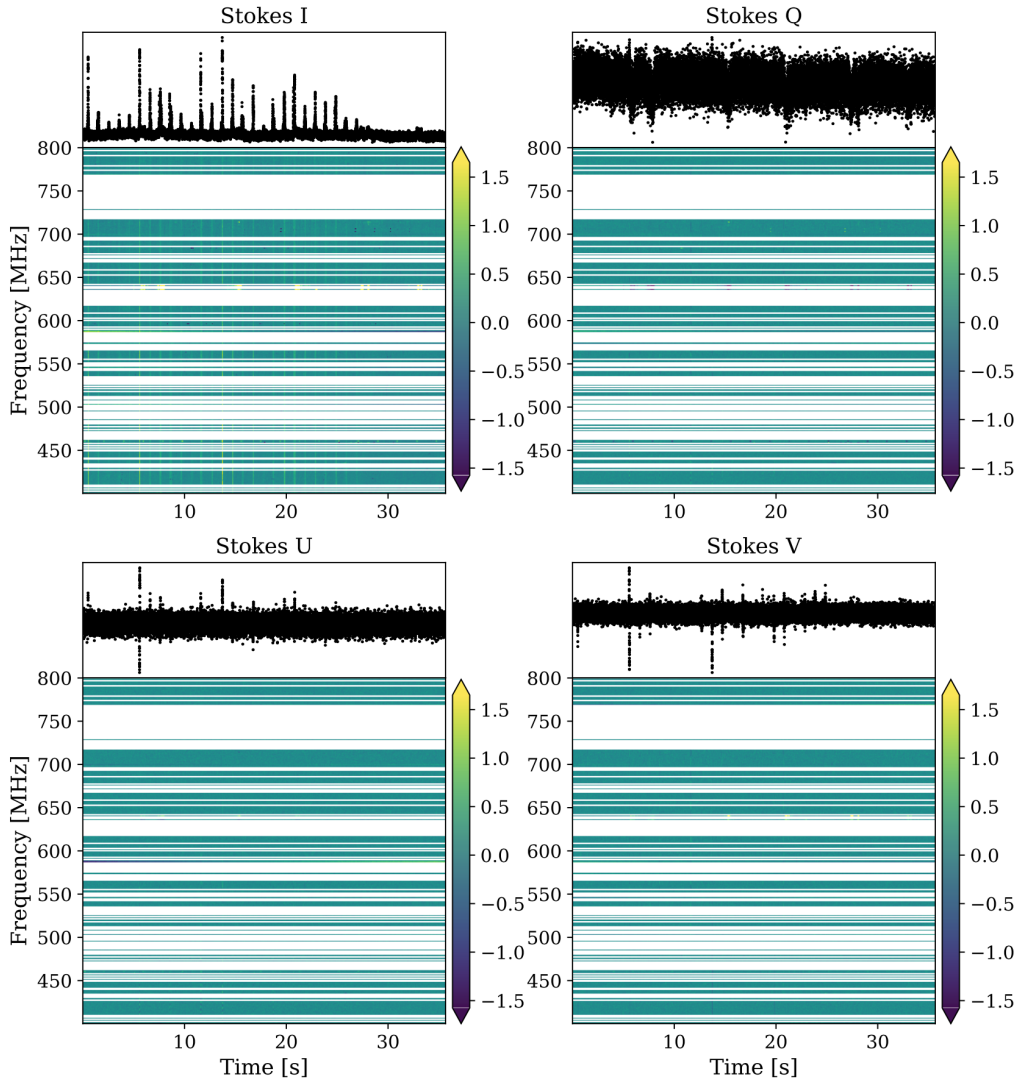
Parameter	Value
Source	2022_CHIME_B2111+46
Observation night	20220826T064420Z
DM (pc cm ⁻³)	141.26
RM (rad m ⁻²)	-218.70
Delay (ns)	-2.0
Start file index	<i>Varied across jobs</i>
Number of files	300
Reference frequency (MHz)	600
Time pixels averaged	391
Slice burst flag	0
$\phi_{\max}$ (rad m ⁻²)	1500
$\Delta\phi$ scaling	0.05
RFI mean threshold	1.7
RFI std threshold	1.3
Search downsampling factor	1
Compute cross-correlation flag	1
Simulation flag	0
Source P0 (s)	1.014684793189
Source W50 (ms)	60.5
FDF downsampling factor	<i>Varied across jobs</i>
Prominence factor	10
Tolerance (ms)	200
Unique data files flag	0

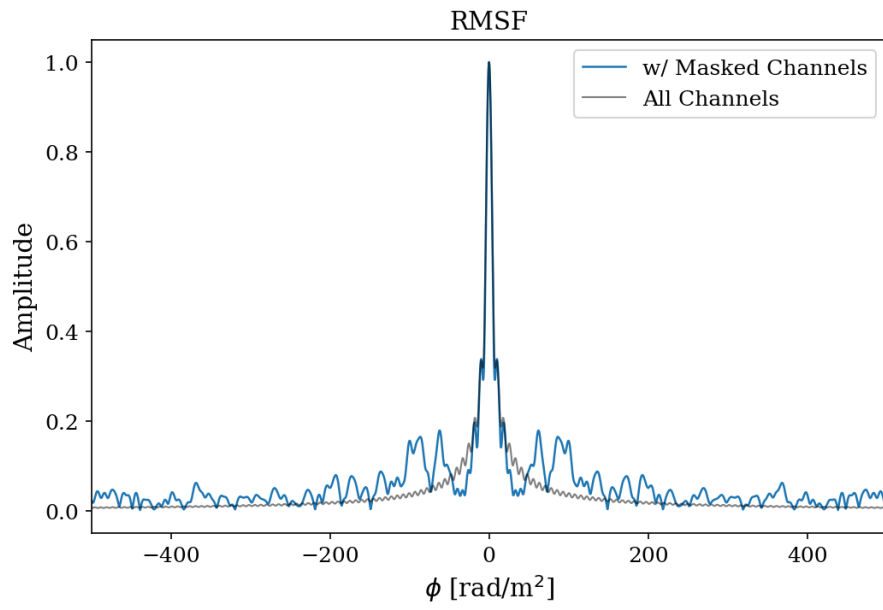
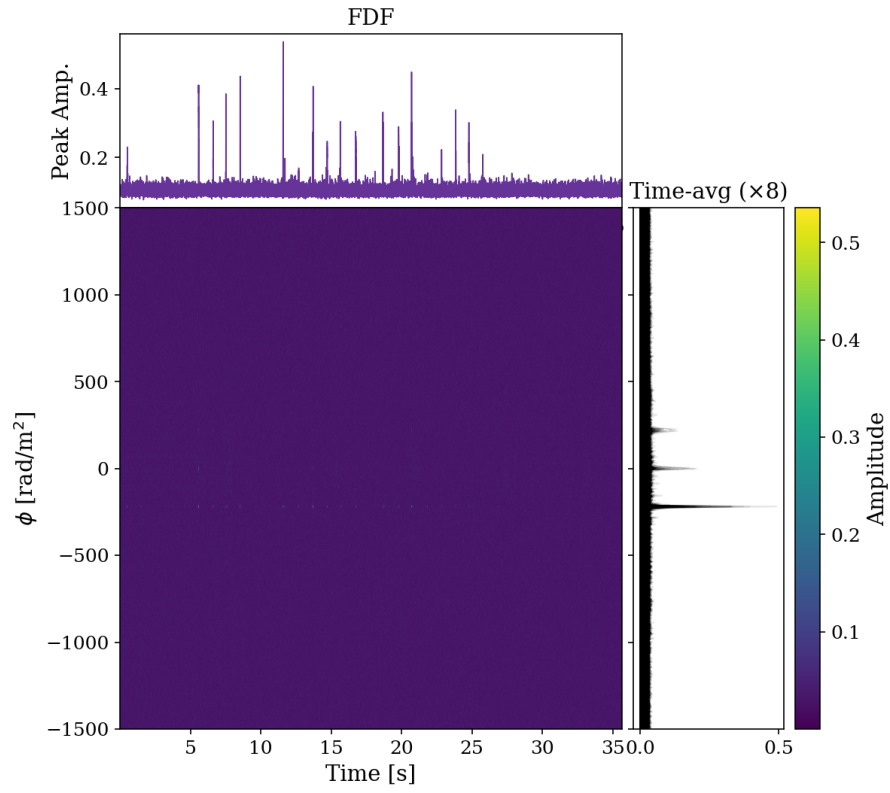
These plots were produced by the scripts 02_pulse_detection.py and 03_plot_search_results.py. I have chosen a specific directory structure to keep track of my results and plots, but this can easily be changed as long as it's kept consistent across scripts. For this reason, I'm not putting here the structure I used, but an example is given in the github repo with the plots from this document in the correct directories.

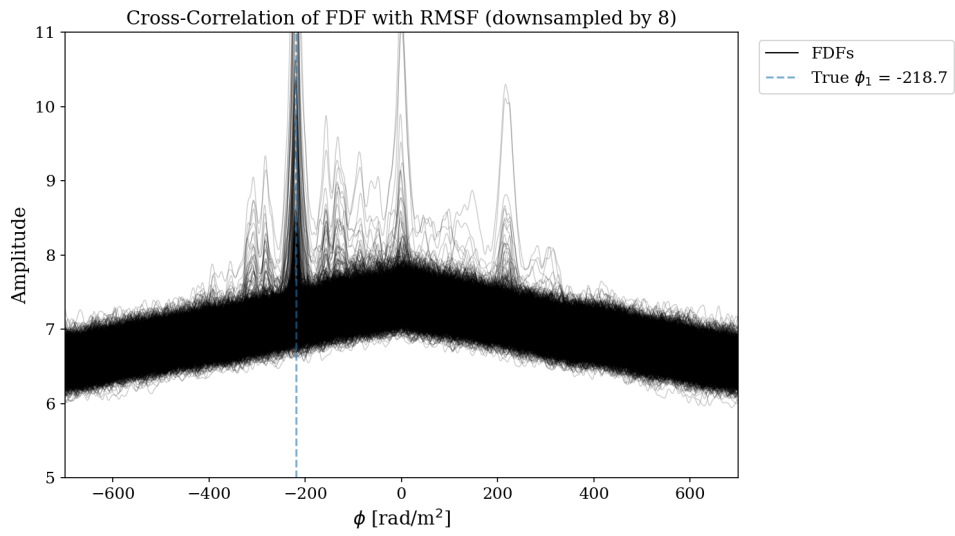
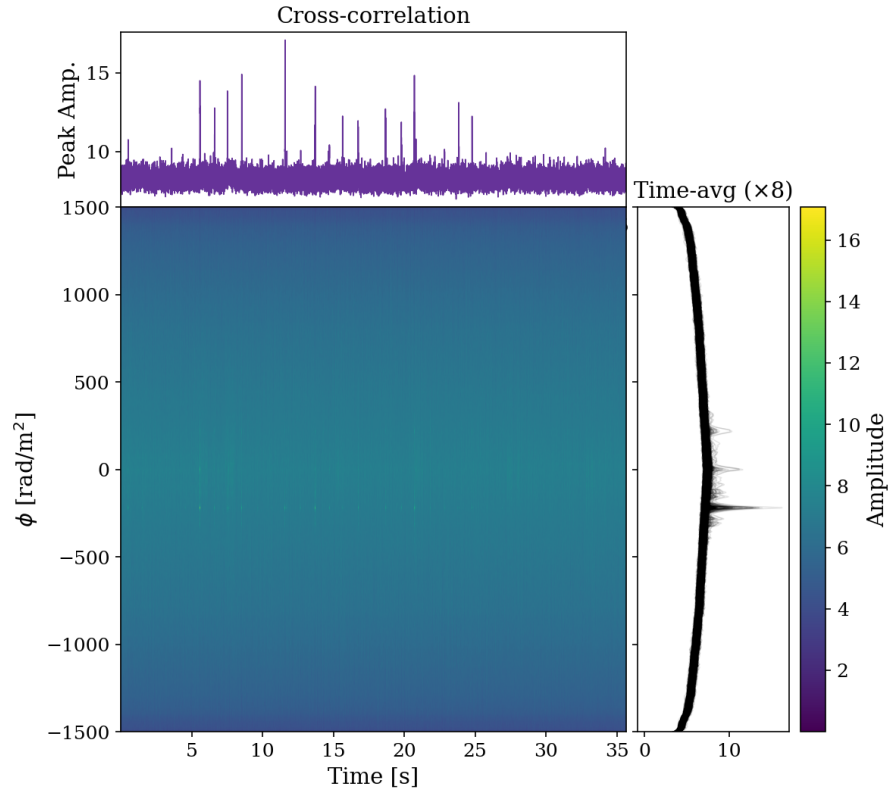
Data and Search Plots

Since the RM search was run as an array job over `start_file_index`, each job processes a different chunk of 300 files and produces one FDF. The following five plots are produced by `03_plot_search_results.py` and correspond to the chunk of data starting at file 2860. These are all saved in the FDFs, stokes, crosscorr, and RMSFs subdirectories of the input `save_fig_dir`.

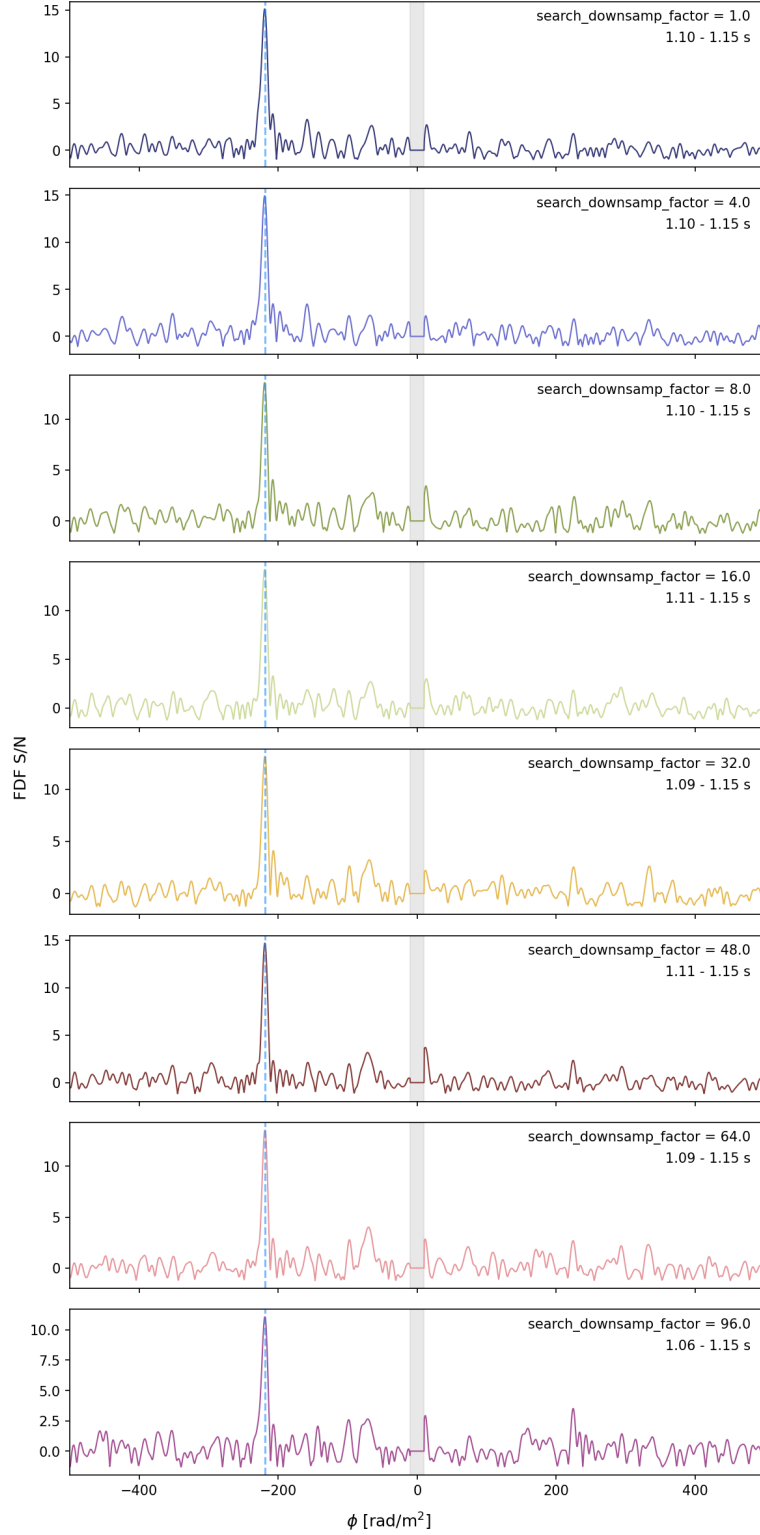
Side note: In the case that the varied parameter does not affect the Stokes data used across runs (eg. if we vary the spacing in ϕ when computing the FDF for a single chunk of data), `03_plot_search_results.py` will produce the Stokes plot only once and then plot all the different FDFs computed during the RM search.

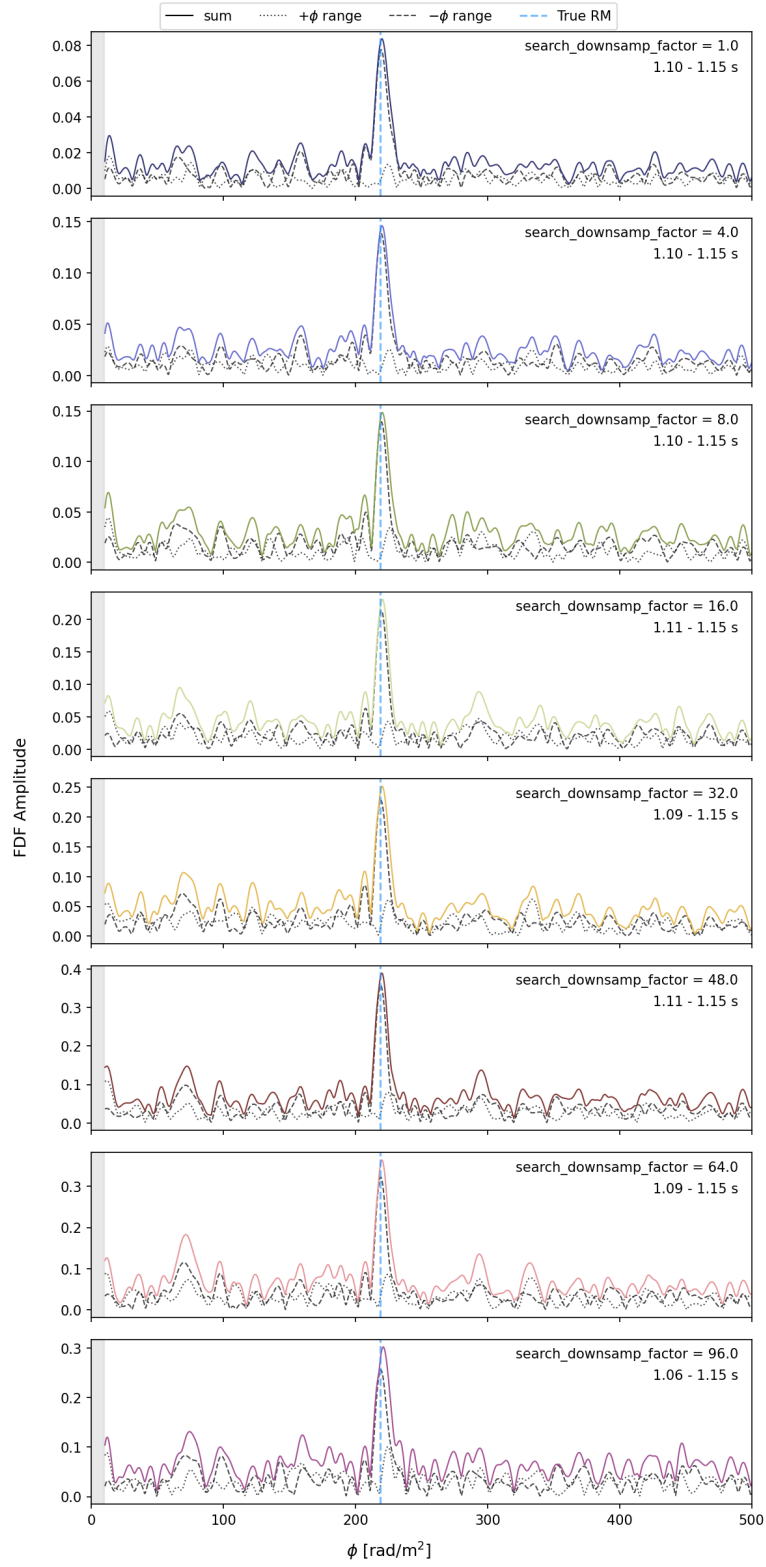


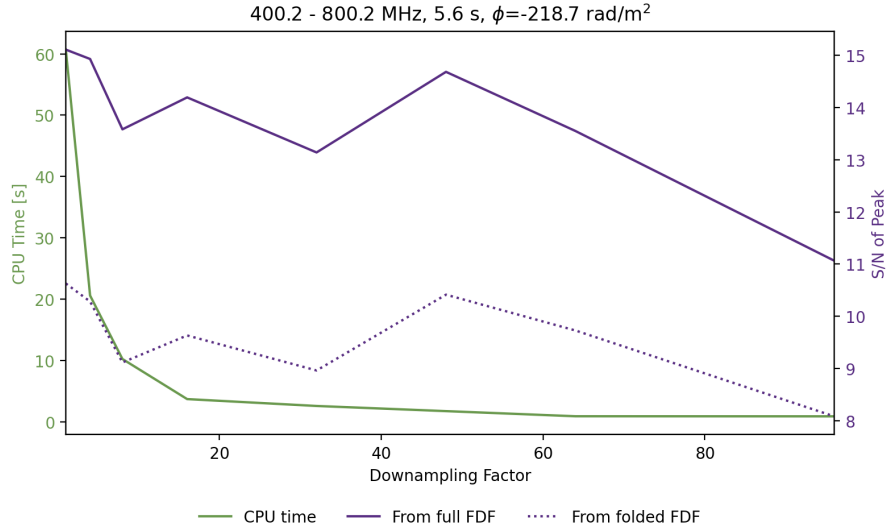




The next three plots are produced in the `single_param_vary_plots.ipynb` notebook rather than by `03_plot_search_results.py`. They require manually selecting an individual burst and its time window to get a plot that looks meaningful. This is used to see how an FDF peak changes as a function of some varied parameter.

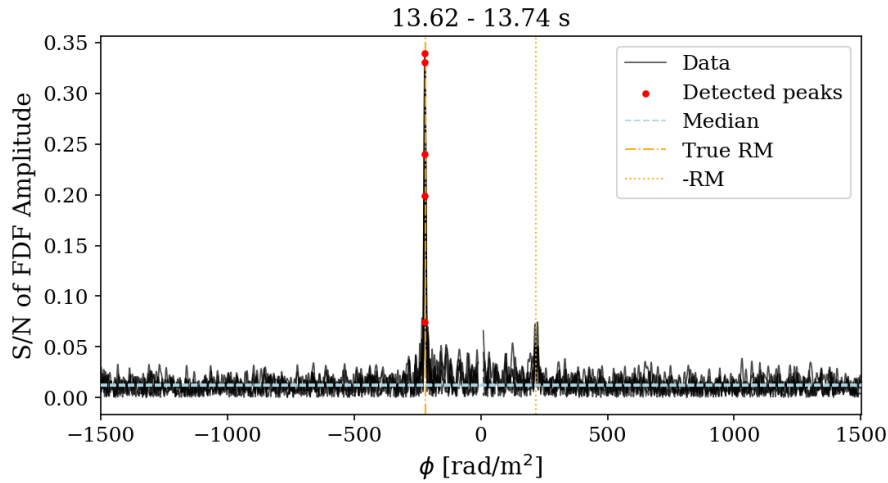






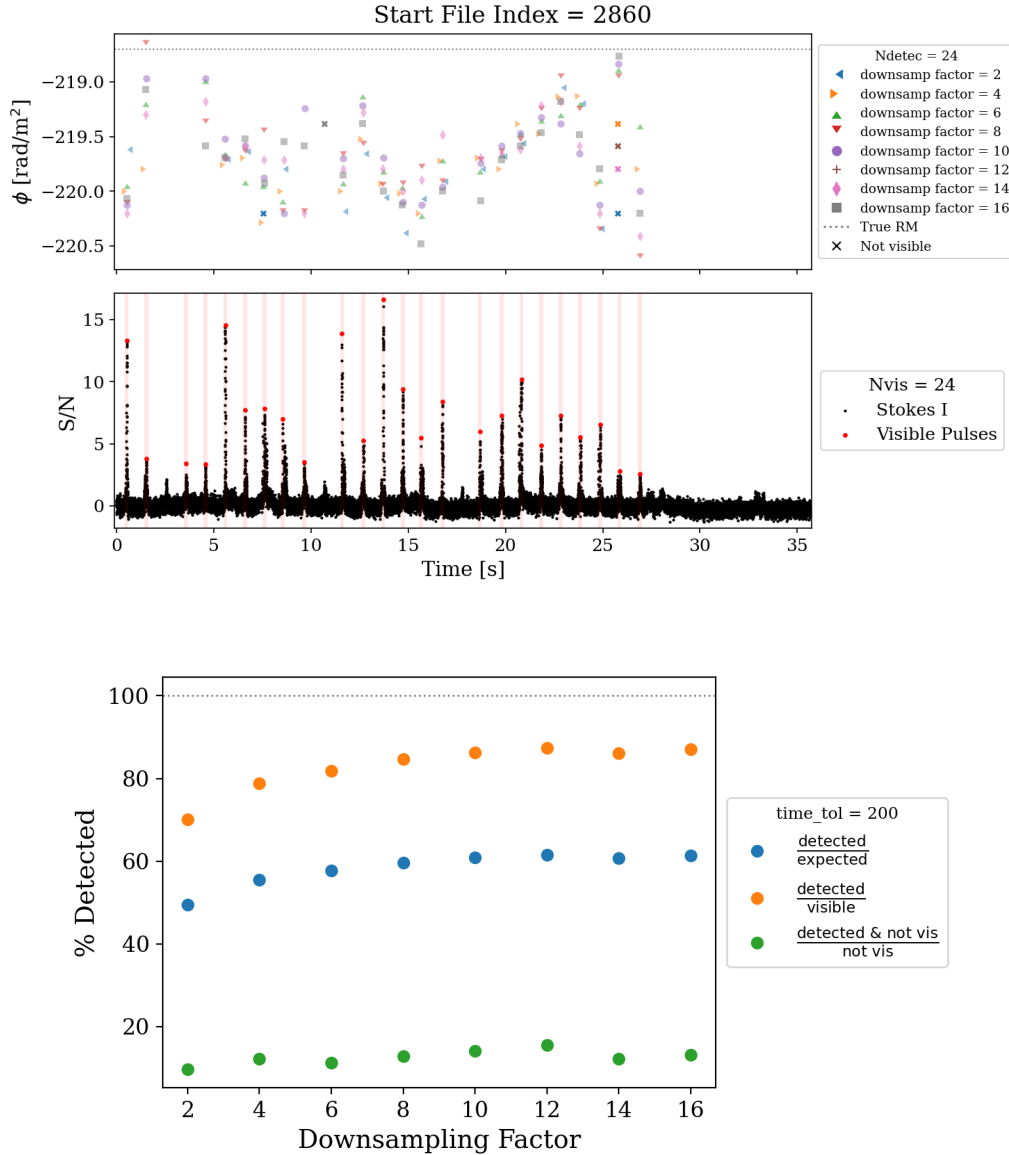
Pulses

A plot like the one below is produced for all pulses found in each FDF (i.e. in each chunk of 300 files) when `02_pulse_detection.py` is run. The code steps through each time bin of the downsampled FDF and searches for peaks along the ϕ axis. Consecutive detections that occur within the specified tolerance are grouped into a single burst. For example, with a tolerance of 200 ms, detections separated by less than 200 ms are treated as belonging to the same burst, otherwise they are counted as 2 different bursts. The plot below shows all the time slices for a single burst found, with their phi peaks in red. This is an easy way to spot if the RM detections are off in phi and/or in time.



The plots below use both the detected and visible pulse information from the results of script 02 and are created by script 03. Visible pulses are identified using a slightly stricter call to `scipy.find_peaks()`, with thresholds on height, prominence, and distance. Since the pulsar period is 1 s, the minimum peak separation is set to 500 ms to avoid double counting visible peaks.

Detected and visible pulses are matched by checking whether a visible peak falls within 50 ms of a detected burst. If this amount or the tolerance for detected peaks are too small, a burst with multiple sub-peaks can be split into two detected bursts. It is then possible that one is matched to the visible pulse while the other is incorrectly labeled as detected but not visible. This can be seen around 26 s in the figure below: the pulse is visible in the bottom panel, but two detections appear in the top panel, with one classified as not visible (x markers).



Timings

These are CPU time and efficiency plots for the different steps in the RM search, also produced by 03_plot_search_results.py

